Early physiological phenotypes and outcomes in critically ill adults with non-traumatic subarachnoid hemorrhage

Take-home message

Among critically ill adults with non-traumatic subarachnoid hemorrhage, eight routine physiological variables collected during the first 48 hours identified three reproducible neuro-systemic phenotypes with graded mortality. The MIMIC-derived classifier preserved this risk gradient in eICU. Early anemia clustered with high-risk phenotypes but was not independently associated with mortality after phenotype adjustment.

Structured Abstract

Purpose

Non-traumatic subarachnoid hemorrhage (NSAH) can trigger systemic organ dysfunction that is not captured by neurological scales alone. We aimed to identify early, transportable physiological phenotypes in critically ill adults with NSAH and to examine their association with mortality and early anemia.

Methods

We conducted a retrospective cohort study using MIMIC-IV 3.1 for derivation and the eICU Collaborative Research Database for external validation. Adult ICU patients with NSAH and ICU stay ≥24 h were included. Eight variables from the first 48 h were selected across neurological, circulatory, oxygenation, renal, hematologic and coagulation domains. Skewed variables were log1p transformed, standardized, reduced by principal component analysis, and clustered using K-means. External validation used frozen MIMIC preprocessing parameters, principal components and centroids.

Results

The derivation cohort included 1,186 patients. Three phenotypes were identified: P1, mild neuro-systemic impairment (n=694); P2, severe neurological impairment with limited systemic dysfunction (n=384); and P3, severe neurological impairment with multisystem shock/organ dysfunction (n=108). In-hospital mortality increased from 6.34% in P1 to 32.55% in P2 and 61.11% in P3. Compared with P1, adjusted odds ratios for death were 7.59 (95% CI 5.07–11.36) for P2 and 21.21 (95% CI 12.08–37.26) for P3. Early anemia was enriched in severe phenotypes but not independently associated with mortality after phenotype adjustment. In eICU (N=843), transported phenotypes preserved the mortality gradient (5.4%, 25.7%, 42.7%).

Conclusions

Routine early ICU physiology identifies transportable NSAH phenotypes with graded mortality risk. These findings support neuro-systemic risk stratification and future phenotype-stratified studies, but not treatment-effect inference.

Keywords: subarachnoid hemorrhage; critical care; phenotyping; unsupervised learning; external validation.

Introduction

Non-traumatic subarachnoid hemorrhage (NSAH) is a high-acuity neurological emergency with substantial mortality and long-term disability. Risk assessment traditionally relies on neurological scales such as Hunt-Hess, WFNS, and the Glasgow Coma Scale (GCS). These tools remain clinically useful, but they incompletely represent the systemic response that frequently follows acute hemorrhage.

Early NSAH may trigger sympathetic activation, systemic inflammation, cardiopulmonary dysfunction, renal hypoperfusion, anemia, and coagulation abnormalities. These extracerebral disturbances can interact with the primary brain injury and may contribute to outcome heterogeneity. General ICU scores such as APACHE and SOFA summarize illness severity, but they were not designed to identify NSAH-specific neuro-systemic patterns.

Unsupervised phenotyping can reveal clinically meaningful subgroups within heterogeneous critical illness syndromes. This approach has informed research in sepsis and acute respiratory distress syndrome, but remains less developed in neurocritical care. We hypothesized that early multimodal physiology would separate critically ill NSAH patients into phenotypes with graded mortality risk; that early anemia would have limited independent prognostic value after accounting for the global phenotype; and that a MIMIC-derived classifier would preserve risk gradients when transported to an external ICU database.

Methods

Study design and data sources

We performed a retrospective cohort study using two de-identified public ICU databases: MIMIC-IV version 3.1 and eICU-CRD version 2.0. MIMIC-IV served as the derivation cohort and eICU-CRD as the external validation cohort. Database access was obtained after completion of Collaborative Institutional Training Initiative training and data-use requirements. Because both resources are de-identified public research databases, consent was waived under source database governance. Additional local ethics documentation should be completed according to the submitting institution and journal requirements. The study follows STROBE guidance; the checklist is intended for Electronic Supplementary Material.

Cohort

Adults admitted to an ICU with NSAH were eligible if ICU length of stay was at least 24 h and no more than two of eight core physiological variables were missing. MIMIC patients were identified using ICD-9 code 430 or ICD-10 I60.x codes. eICU patients were identified using ICD-9 code 430, admission diagnosis text containing subarachnoid hemorrhage, or diagnosis table entries consistent with NSAH. Patients with traumatic subarachnoid hemorrhage, multiple ICU stays during the index hospitalization, or massive red blood cell transfusion within 24 h (operationally ≥5 units) were excluded. The latter exclusion was used to reduce distortion of baseline hemoglobin, hemodynamics and coagulation signals.

Variables and preprocessing

Eight variables were selected a priori from the first 48 h after ICU admission: minimum GCS motor score, minimum mean arterial pressure, maximum shock index, minimum oxygen saturation, maximum creatinine, maximum international normalized ratio, minimum hemoglobin, and minimum platelet count. Values were range-filtered using clinically plausible limits. Missing values were imputed using derivation-cohort medians. Creatinine and international normalized ratio were log1p transformed because of right-skewed distributions. All variables were standardized with derivation-cohort means and standard deviations.

Phenotype derivation and validation

Principal component analysis was applied to the standardized eight-variable matrix. Three components were retained, explaining 56.41% of variance. K-means clustering was performed in the three-component space using `random_state = 42` and `n_init = 100`. K=3 was selected based on cluster metrics, minimum cluster size, bootstrap stability, and clinical interpretability. Phenotypes were ordered by increasing mortality.

External validation used frozen transport. eICU data were imputed, transformed and standardized with MIMIC parameters, projected with MIMIC principal component loadings, and assigned to the nearest MIMIC phenotype centroid. Transported phenotypes were evaluated against mortality and independent eICU severity measures. De novo eICU clustering was used only as structural sensitivity analysis.

Outcomes and statistical analysis

The primary outcome was in-hospital mortality. Secondary outcomes included ICU mortality, length of stay, early anemia, and red blood cell transfusion. Multivariable logistic regression assessed the association between phenotype and hospital mortality. The primary model adjusted for age, sex, admission type, NSAH evidence level, aneurysm diagnosis, and early anemia. A process-of-care model additionally included nimodipine, vasopressors, mechanical ventilation, red blood cell transfusion, renal replacement therapy, EVD/ICP monitoring, and fluid balance. Process variables were interpreted as exploratory severity and treatment-selection markers, not causal treatment effects. Cox models were used as sensitivity analyses for time-to-death during hospitalization. Sensitivity analyses included complete-case, strict aneurysm, ICU stay ≥48 h, 0–24 h window, hemoglobin-free, INR-free, K=4, and 200-bootstrap stability analyses.

Results

Cohort and phenotype structure

The MIMIC derivation cohort included 1,186 adults. Overall in-hospital mortality was 19.81%, early anemia occurred in 26.56%, and red blood cell transfusion within 48 h occurred in 2.02%. Missingness was low in the derivation cohort, with maximum INR missing in 5.48% and all other core features missing in ≤0.08%.

The K=3 solution identified three clinically interpretable phenotypes (Fig. 1 and Fig. 2). P1 included 694 patients (58.5%) and was characterized by mild neurological and systemic impairment. P2 included 384 patients (32.4%) and showed severe neurological impairment with relatively preserved systemic physiology. P3 included 108 patients (9.1%) and combined severe neurological impairment with hypotension, elevated shock index, hypoxemia, renal dysfunction, coagulopathy, thrombocytopenia, and lower hemoglobin.

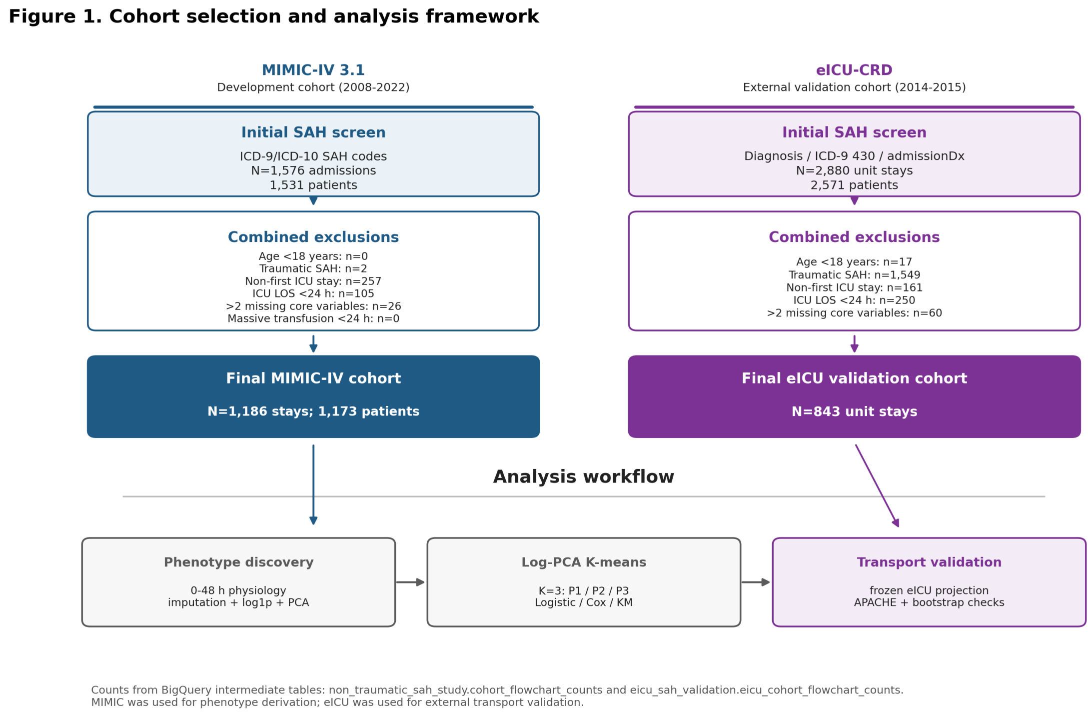

Fig. 1. Cohort selection and analysis framework. Counts were derived from BigQuery intermediate cohort-flow tables. The lower panel summarizes MIMIC phenotype derivation and eICU frozen transport.

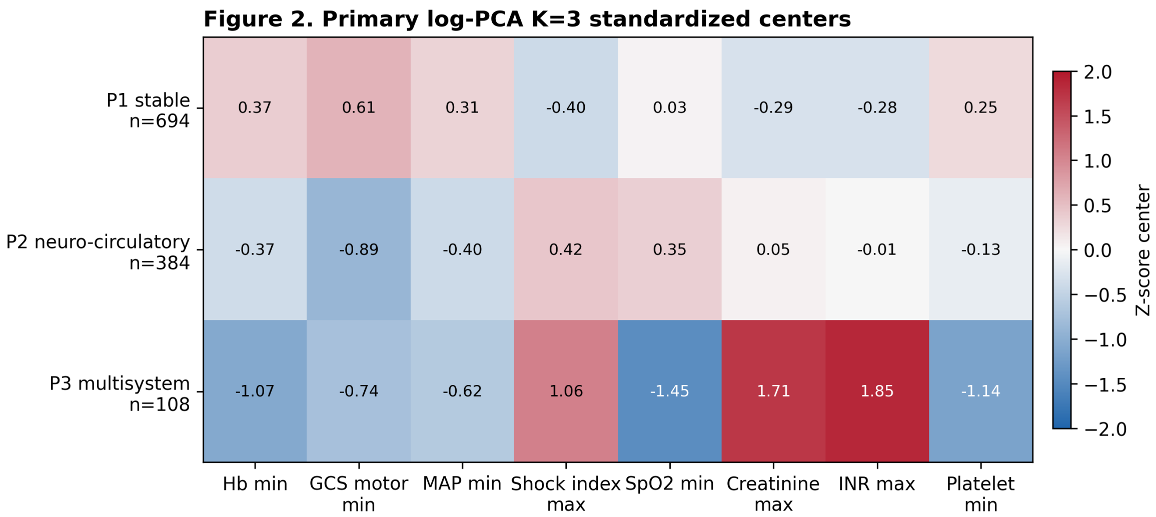

Fig. 2. Early physiological profiles of the three MIMIC-derived phenotypes. Values represent standardized cluster centers with raw medians and interquartile ranges.

Outcomes and anemia

Mortality increased monotonically across phenotypes (Fig. 3). In-hospital mortality was 6.34% in P1, 32.55% in P2, and 61.11% in P3. ICU mortality followed the same pattern: 3.60%, 26.56%, and 50.93%, respectively. In unadjusted Cox analysis, hazard ratios for hospital death were 4.20 (95% CI 2.97–5.94) for P2 and 7.94 (95% CI 5.38–11.70) for P3 compared with P1.

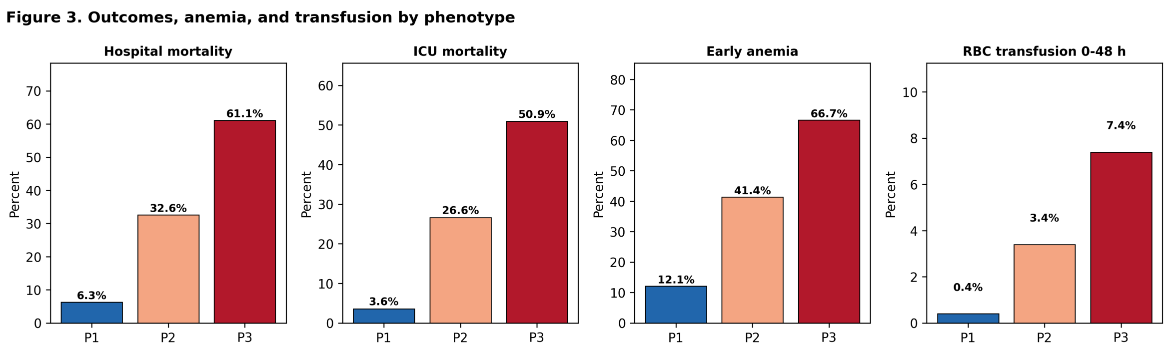

Fig. 3. Outcome, anemia and early red blood cell transfusion gradients in MIMIC-IV. Red blood cell transfusion is shown as a descriptive process variable and should not be interpreted as a treatment-effect estimate.

In the primary logistic model, P2 and P3 remained associated with in-hospital mortality after adjustment. Compared with P1, adjusted odds ratios were 7.59 (95% CI 5.07–11.36) for P2 and 21.21 (95% CI 12.08–37.26) for P3. Early anemia was enriched in P2 and P3 but was not independently associated with mortality after phenotype adjustment (adjusted odds ratio 0.99, 95% CI 0.68–1.44). The process-of-care model attenuated but did not remove phenotype associations. These process-adjusted estimates are exploratory because several variables occurred during the feature window.

External validation

The eICU validation cohort included 843 patients. Frozen transport assigned 539 patients to P1, 222 to P2, and 82 to P3. Hospital mortality increased from 5.4% in P1 to 25.7% in P2 and 42.7% in P3. ICU mortality, early anemia, and red blood cell transfusion also increased across transported phenotypes.

Transported phenotype order aligned with independent eICU severity measures (Fig. 4). Median APACHE scores were 36, 57, and 79 across P1, P2, and P3, with Spearman rho 0.480. Acute Physiology Score and predicted hospital mortality showed similar gradients. These variables were not used for phenotype assignment.

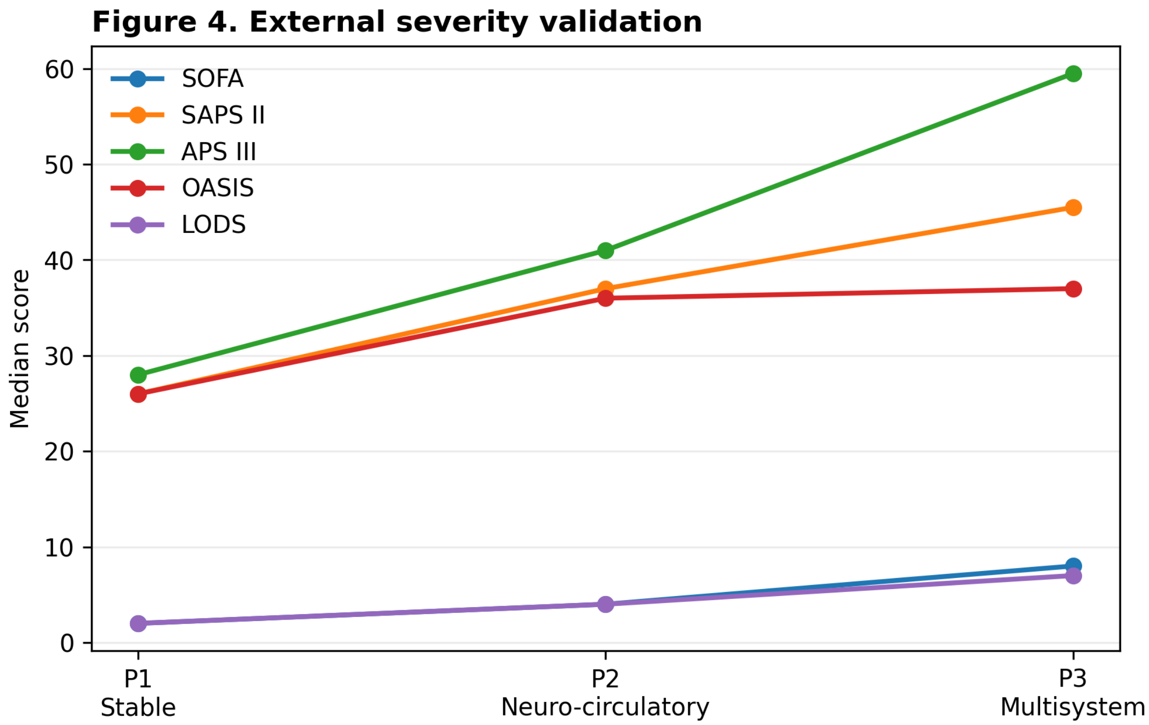

Fig. 4. External criterion validation in eICU. APACHE, Acute Physiology Score and predicted mortality were not clustering inputs.

Independent de novo K-means clustering in eICU recovered a mortality gradient, but patient-level agreement with transported labels was low (adjusted Rand index -0.003). This supports transportability of risk gradients, not exact replication of cluster boundaries across databases.

Prediction and robustness

A multivariable eight-variable physiological model improved mortality prediction compared with GCS alone (Fig. 5). Cross-validated AUROC was 0.842 for the full physiological model, 0.754 for phenotype only, and 0.539 for GCS only. SHAP-style attribution ranked GCS motor, creatinine, and platelets as the most influential variables, consistent with neurological, renal and coagulation axes.

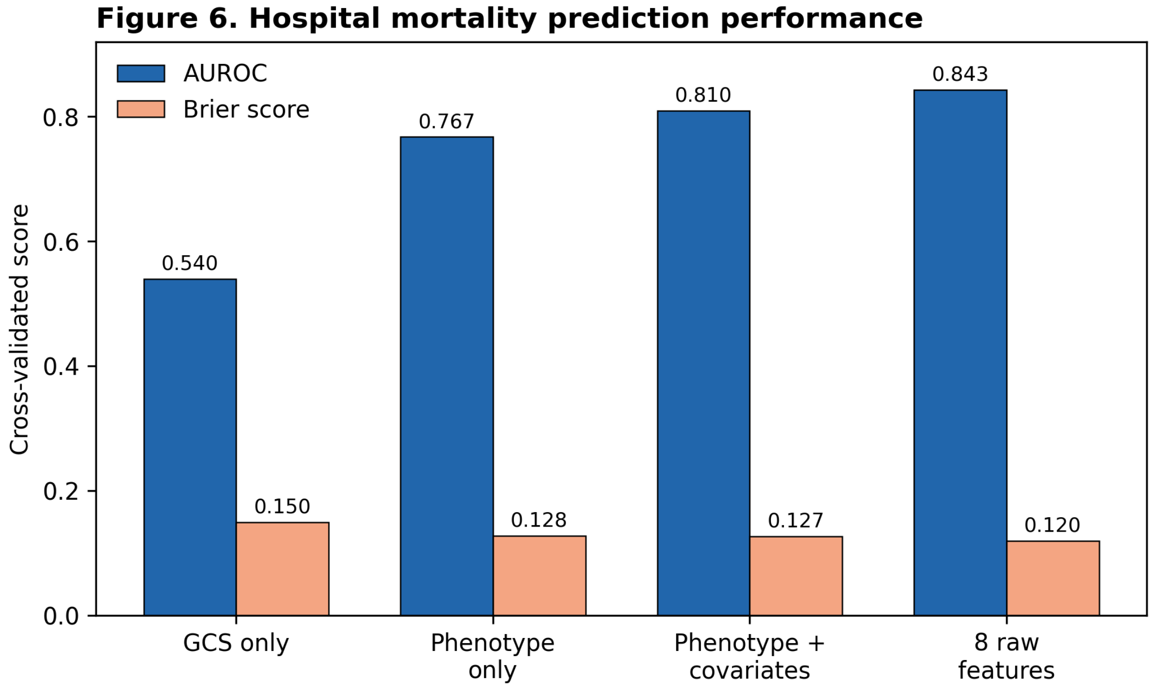

Fig. 5. Incremental prediction of in-hospital mortality in MIMIC-IV.

Sensitivity analyses preserved the ordered mortality gradient. Bootstrap stability was high, with mean adjusted Rand index 0.920 across 200 resamples. Hemoglobin-free clustering preserved phenotype-mortality separation and supported the conclusion that the anemia finding was not solely due to including hemoglobin in the primary clustering model.

Discussion

This study identified three early NSAH phenotypes using routine ICU physiology and validated their risk gradient in an external multicenter database. The phenotypes separated patients with mild neuro-systemic impairment, severe neurological impairment with limited systemic dysfunction, and severe neurological impairment with multisystem organ dysfunction. The distinction between P2 and P3 is clinically relevant: both had severe neurological impairment, but only P3 showed broad systemic derangement and markedly higher mortality.

These findings complement existing scales rather than replacing them. Hunt-Hess, WFNS and GCS emphasize neurological status. APACHE and SOFA summarize general critical illness severity. The present phenotypes occupy an intermediate space: they retain neurological interpretability while capturing systemic organ responses that are common after NSAH. This may be useful for prognosis, cohort enrichment, and future trials in which heterogeneous systemic physiology can obscure treatment signals.

The anemia results require cautious interpretation. Early anemia was common in high-risk phenotypes but was not independently associated with mortality after phenotype adjustment. This suggests that anemia may function as a marker of global physiological derangement rather than an isolated prognostic driver. The observation is consistent with recent randomized evidence in aneurysmal subarachnoid hemorrhage showing no clear long-term neurological benefit from a liberal transfusion strategy. However, our study does not estimate transfusion effects and should not guide transfusion thresholds. Phenotype-stratified causal analyses or randomized trials would be needed.

External validation showed preserved mortality and severity gradients in eICU. At the same time, de novo eICU clustering did not reproduce patient-level labels. This apparent tension is important. It suggests that the risk-stratifying structure is transportable, whereas exact mathematical boundaries are sensitive to database-specific measurement, missingness and documentation patterns. For clinical translation, frozen classifiers may therefore be preferable to expecting identical clusters to emerge independently in every dataset.

The main strength of this study is the use of a simple, reproducible pipeline based on routine variables available early in ICU care, with external validation and multiple sensitivity analyses. The approach is deliberately conservative: it avoids high-dimensional black-box modeling for phenotype derivation and focuses on interpretable neuro-systemic patterns.

Several limitations should be acknowledged. First, MIMIC-IV and eICU are retrospective observational databases, so residual confounding and misclassification remain possible. Second, detailed neuroimaging variables, including Fisher grade, hemorrhage volume, aneurysm location, hydrocephalus and procedure timing, were unavailable. Third, the 0–48 h window may include physiology modified by early ICU interventions. Fourth, eICU INR missingness was substantial, although INR-free sensitivity analyses preserved risk gradients. Finally, both databases are from US healthcare settings; prospective validation in contemporary and geographically diverse neurocritical care cohorts is required.

Conclusions

Eight routine physiological variables measured during the first 48 h of ICU admission identified three NSAH phenotypes with graded mortality risk. The MIMIC-derived classifier preserved this gradient in eICU and aligned with independent severity measures. Early anemia appeared to mark severe systemic derangement rather than independently predicting mortality after phenotype adjustment. These phenotypes may support risk stratification and future trial design, pending prospective validation.

Declarations

Funding: To be completed by authors before submission.

Conflicts of interest: To be completed by authors before submission.

Ethics approval: MIMIC-IV and eICU-CRD are publicly available de-identified research databases. Database access was obtained after required training and data-use agreements. Local ethics documentation should be completed according to the submitting institution and target journal requirements.

Consent to participate: Waived under the source database governance for de-identified public research resources.

Data availability: MIMIC-IV and eICU-CRD are available through PhysioNet to credentialed users who complete required training and data-use agreements. Derived aggregate outputs are included in the manuscript and Electronic Supplementary Material.

Code availability: Analysis code and reproducible preprocessing parameters should be provided in a public repository or submitted as supplementary material before journal submission.

Author contributions: To be completed by authors before submission.

Use of AI-assisted tools: AI-assisted drafting and editing tools were used to help prepare manuscript text and formatting. All scientific content, analyses, data interpretation and final wording require author verification before submission.

Reporting guideline: The STROBE checklist will be submitted as Electronic Supplementary Material.

Electronic Supplementary Material

The Electronic Supplementary Material is prepared as `electronic_supplementary_material.md` and includes the cohort algorithm, ICD/code-list, variable mapping, missingness audit, extended baseline/phenotype/regression/Cox/eICU/sensitivity tables, supplementary figures, BigQuery provenance, and reproducibility parameters. The STROBE checklist is prepared separately as `strobe_checklist.md`.

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